

Evaluating an LLM-Based Phenotype Extractor from Medical Reports Using Combinatorial Testing

This is the replication package for the paper "Evaluating an LLM-Based Phenotype Extractor from Medical Reports Using Combinatorial Testing".

The LLM-based phenotype extractor is available at <https://doi.org/10.5281/zenodo.17912782>, including all configuration files, scripts, and instructions on how to set up and use the extractor.

Requirements for this Replication Package

Before running the scripts, install the following dependencies:

For the Majority of programs operating with csv:

```
pip install pandas
```

For translation:

```
pip install deep_translator
```

For scripts operating on the HPO ontology:

```
pip install pronto
```

The HPO ontology, contained into the file **hp.obo**, is also required, and must be stored under the **utils** directory.

You can download it from the following link: <https://hpo.jax.org/data/ontology>

Usage

Folder Structure

- The folder contains Python scripts that process and analyze data.
- The CSV files are located in the **output** folder.
- The **utils** folder contains utility python scripts.

Dataset Preparation

This replication package already contains the resulting datasets (**merged_dataset.csv** and **merged_dataset_with_parents.csv**). Thus, users who don't need to translate or merge datasets can skip this section.

- **save_and_prepare_database.py**: It saves the dataset needed for further operations. Then, it translates a specific dataset (translates 10 rows to English per cycle), and merges the two datasets into

a single file. It also customizes the dataset by adding different names for patients, physicians, and hospitals.

After running this program, you will have a dataset named `merged_dataset.csv` and one named `merged_dataset_with_parents.csv` containing also parent codes.

The functionalities of `save_and_prepare_database.py` exploits the file `utils/names.py` which contains the names of patients, physicians, and hospitals.

IPM Creation

- `generate_ipm.py`: It generates the IPM, for a given depth and number of parameters. The IPM is saved in a file named `test.ctw`, in CTWedge format, and `hpo.acts` in the format compatible with CAgem.

Combinatorial sampling

- `combinatorial_sampling.py`: It generates the file `2wise_test_suite.csv` which contains the real test cases to be executed against the LLM.

**** Aggiornato fino a qua****

HPO Hierarchy

- `completehierarchy_from118.py`: Organizes the HPO hierarchy with a maximum parent code of `HP:0000118` into a CSV file.

CTWedge Execution

- Execute CTWedge on the generated input files and save the result as `code.csv`.

Code Formatting

- `codectwformatting.py`: Formats the codes by converting the abbreviated form `HP:0000001` in the `code.csv` file (input required).

Extraction of Matching Combinations

- `addingparentstodataset.py`: Adds parent codes to their children in the ordered dataset and stores the result in a new column.

Association

- `csvcreator_finale.py`: Matches the codes from the CTWedge combination with all the corresponding reports in your dataset.
- `find_best_match.py`:Matches the code from the CTWedge combination with the best corresponding report in your dataset.
- `**from_ctw_to_acts.py**`translate from ctw to acts specifically for ctw hpo

Input & Outputs

Script name	Inputs	Outputs
save_and_prepare_database.py		merged_dataset.csv

Script name	Inputs	Outputs
completehierarchy.py	hp.obo	completehpohierarchy.csv
completehierarchy_from118.py	hp.obo	completehpohierarchy_from118.csv
depthfrom118.py	completehpohierarchy.csv	true_depth_from118.csv
CTWEDGE EXECUTION	test.ctw	codes.csv
codectwformatting.py	code.csv	formatted_code.csv
find_best_match.py	formatted_codes.csv	output_data_matches
	ordered_dataset_withparents.csv	output_data_no_matches
	completehpohierarchy.csv	output_data_exact_matches

Contributing

Pull requests are welcome! For major changes, please open an issue first to discuss the proposed changes.

Please ensure to update the tests as appropriate.

License

[Insert License Information Here]